



AGROCARE: PLANT DISEASE DETECTION & CROP RECOMMENDATION SYSTEM A PROJECT REPORT

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ABSTRACT

Agriculture remains the backbone of food security and economic stability, especially in developing regions where farming supports the majority of livelihoods. To address challenges such as crop diseases, unpredictable weather, and inefficient resource use, AgroCare was developed as an AI-driven agricultural assistance system integrating Plant Disease Detection and Crop Recommendation modules. The system combines deep learning, machine learning, and IoT technologies to deliver real-time, data-driven insights that enhance productivity and promote sustainable farming practices.

During the seventh semester, major improvements were made to both modules. The Plant Disease Detection component underwent extensive software-level enhancements. Its architecture was optimized using advanced Convolutional Neural Networks and transfer learning, which improved feature extraction, reduced training time, and increased accuracy across a wider variety of plant species and diseases. These upgrades strengthened the model's reliability and scalability across diverse agricultural conditions. No hardware integration was included during this phase, as the focus remained on refining algorithms and expanding the dataset.

In parallel, the Crop Recommendation System was expanded using both hardware-based and API-driven data acquisition. IoT-enabled sensing through Raspberry Pi and Arduino allowed real-time measurement of essential soil parameters such as pH, nitrogen, phosphorus, and potassium. To complement this, a weather API module was integrated to fetch dynamic environmental data, including temperature, humidity, and rainfall. This hybrid input approach enables the recommendation model to adapt intelligently to local and seasonal conditions, resulting in more accurate and region-specific crop suggestions.

Together, these advancements represent a significant step toward precision agriculture, where continuous monitoring and intelligent analysis guide decision-making. AgroCare now provides farmers with more accurate insights, reduces reliance on manual assessment, and supports sustainable resource management. Future work will focus on field deployment, real-world data validation, and expanding accessibility through low-cost devices and multilingual support. The project remains open-source on GitHub, encouraging global collaboration to further improve its scalability and performance.

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1. INTRODUCTION

Agriculture plays a pivotal role in sustaining economies, providing food security, employment, and contributing to national income, especially in developing countries. Despite advancements in farming methods, farmers continue to face several critical challenges, including plant diseases, inefficient crop management, soil degradation, and the unpredictable impacts of climate change. Addressing these challenges requires a shift from traditional practices to modern, technology-driven solutions.

Recent developments in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and the Internet of Things (IoT) have opened new avenues for transforming agriculture. Predictive analytics, real-time monitoring, and data-driven recommendations are now increasingly accessible to farmers, thanks to advances in computational power and sensor technology. These technologies enable early detection of plant diseases, optimization of crop choices, efficient resource management, and overall improvement in agricultural productivity.

AgroCare is a proposed AI-powered system that integrates Plant Disease Detection and Crop Recommendation into a unified platform. By using Convolutional Neural Networks (CNNs) for image-based disease diagnosis and machine learning algorithms for crop recommendations based on soil and environmental parameters, AgroCare empowers farmers with actionable insights in real time. The system leverages edge computing devices like Raspberry Pi and NVIDIA Jetson Nano along with IoT-based sensors to enable continuous monitoring and on-site decision-making without heavy cloud dependency.

2. LITERATURE REVIEW

In recent years, artificial intelligence and data-driven techniques have gained significant importance in agriculture, with numerous studies focusing on plant disease detection and crop recommendation. These systems aim to enhance productivity, reduce crop losses, and support farmers with accurate, data-based decision-making.

Early plant disease detection research relied on traditional machine learning methods such as SVM and Random Forest, using manually extracted features from leaf images (Patil et al., 2019; Singh et al., 2020). However, these models showed limited accuracy under varying environmental conditions. With advancements in deep learning, Convolutional Neural Networks (CNNs) demonstrated superior performance by automatically learning features and achieving high accuracy in disease classification, as highlighted by Ferentinos (2018) with results exceeding 95%.

For crop recommendation, earlier studies like Akinbile et al. (2021) utilized soil parameters (NPK, pH) to suggest suitable crops, while others incorporated climatic factors such as temperature and rainfall for better reliability. IoT-based systems proposed by Rajeswari et al. (2022) enabled real-time soil monitoring but lacked integration of external weather data, limiting region-specific adaptability.

A review of existing research indicates that most solutions address either plant disease detection or crop recommendation independently. Only a few combine both, and even fewer integrate real-time soil sensing hardware with dynamic weather data from APIs. The absence of such hybrid systems creates a gap in achieving a more comprehensive and adaptive agricultural support tool.

To address these limitations, the proposed project integrates hardware-based soil sensing with API-driven environmental data to enhance both plant disease detection and crop recommendation. By combining real-time soil information with dynamic temperature, humidity, and rainfall inputs, the system is designed to deliver a more accurate, automated, and reliable smart agriculture solution capable of supporting farmers across diverse agricultural conditions.

3. SCHEME AND IMPLEMENTATION

3.1 System Architecture Overview

The AgroCare system is designed as a modular architecture consisting of two major components:

- (1) Plant Disease Detection Module and
- (2) Crop Recommendation Module.

Each module follows an independent workflow but both culminate in providing decision-support to farmers through an integrated interface. The system combines machine learning models, image processing techniques, hardware-based soil sensing, and API-driven weather data to ensure accurate and real-time agricultural insights.

The architecture follows a three-tier structured scheme:

- **Data Acquisition Layer** – Responsible for collecting soil data (via sensors), environmental data (via API), and plant images (via user upload or dataset).
- **Processing and Analysis Layer** – Implements deep learning and machine learning algorithms for disease classification and crop prediction.
- **Application Layer** – Web interface providing user interaction, data submission, and display of predictions.

4. DATASET PREPARATION

4.1 Crop Recommendation System

4.1.1 Objective:

The second module of the project focuses on assisting farmers in making data-driven decisions regarding crop selection based on real-time environmental and soil conditions. By integrating machine learning techniques with sensor-based soil data and live weather information, the system aims to optimize land utilization, improve crop yield, and promote sustainable farming practices.

4.1.2 Dataset Overview

Dataset Source: Kaggle - Crop Recommendation Dataset

Link- <https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>

Link- <https://www.kaggle.com/datasets/shankarpriya2913/crop-and-soil-dataset>

These datasets provide comprehensive information on soil nutrients and environmental parameters, suitable for training machine learning models in precision agriculture.

4.1.3 Dataset Description

The dataset integrates multiple sources of agricultural data, including climate, soil nutrients, and rainfall statistics, to determine the most appropriate crops for a given set of conditions. It was curated to reflect Indian agricultural environments, making it ideal for region-specific crop prediction.

The dataset was further enhanced with real-time inputs obtained from:

- Soil sensors that measure parameters such as soil moisture, soil temperature, and pH.
- A Weather API that provides live data on temperature, humidity, and rainfall.

This integration allows the system to dynamically recommend crops based on current field conditions rather than relying solely on static datasets.

4.1.4 Key Variables

Table 1. Key Variables and their descriptions

Variable	Description
N	Nitrogen content ratio in the soil
P	Phosphorus content ratio in the soil
K	Potassium content ratio in the soil
Temperature	Average temperature in degrees Celsius
Humidity	Relative humidity
pH	Soil pH level
Rainfall	Average annual rainfall (in millimeters)
Soil Temperature	Temperature of soil (°C)
Soil Moisture	Moisture content of soil (%)

4.1.5 Importance of the Parameters:

- **Nitrogen (N):** Essential for leaf growth and chlorophyll formation in plants.
- **Phosphorus (P):** Promotes root development and improves flowering and fruiting.
- **Potassium (K):** Enhances overall plant health and disease resistance.
- **Temperature:** Affects seed germination, growth rate, and crop yield.
- **Humidity:** Influences transpiration rate and water availability to plants.
- **pH:** Determines soil acidity or alkalinity, affecting nutrient absorption.
- **Rainfall:** Provides necessary water for crop growth and soil moisture balance.
- **Soil Temperature:** Impacts microbial activity and nutrient availability in the soil.
- **Soil Moisture:** Maintains hydration levels crucial for seed germination and plant growth.

4.1.6 Use Case Scenario:

In this system, hardware sensors provide real-time soil data such as soil temperature, soil moisture, and pH, while an API retrieves weather parameters including rainfall, humidity, and temperature. Along with soil nutrient values (N, P, K), these inputs are processed by the machine learning model to recommend the most suitable crop for the given conditions. This combination of hardware sensing and software prediction ensures accurate, location-specific recommendations, enabling farmers to make informed and efficient agricultural decisions.

Model Info:

- **Type:** Supervised classification
- **Algorithms Used:** Decision Tree, Random Forest, KNN, XGBoost
- **Input:** 9 numerical features
- **Output:** Predicted crop label (e.g., rice, wheat, maize, muskmelon, etc.)

5. WORKFLOW DIAGRAM

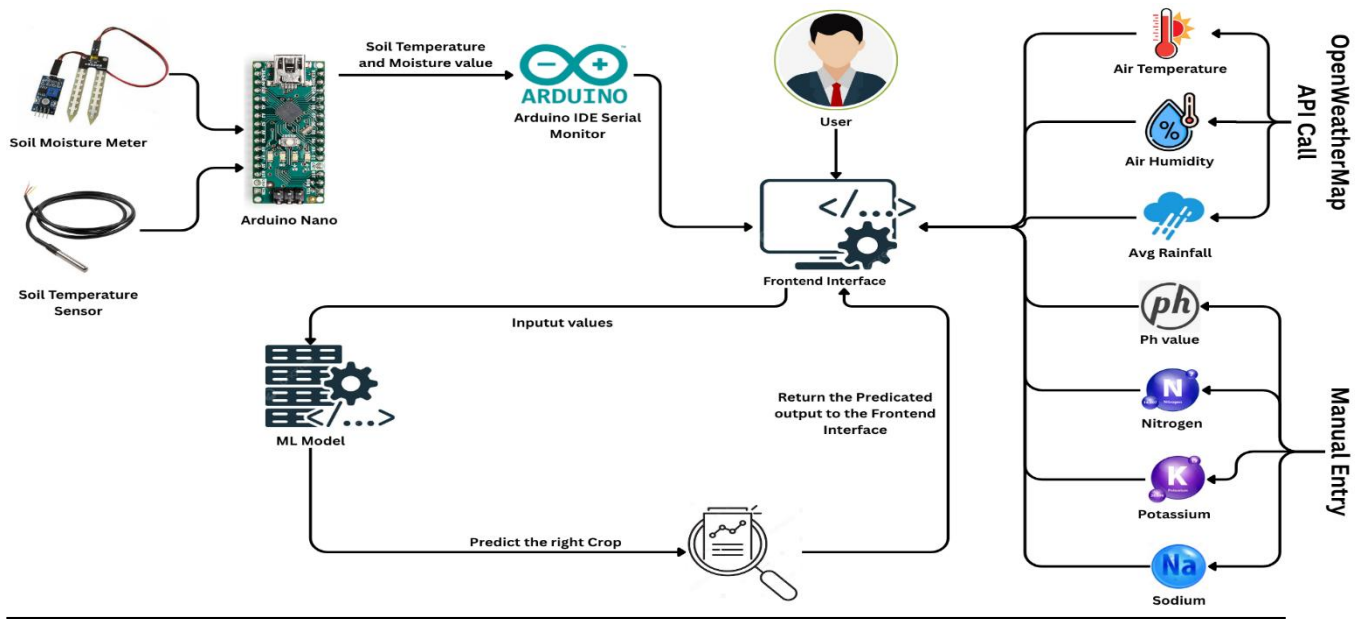


Fig 1. Workflow diagram

The flow diagram (Fig 1.) illustrates how different components work together in the Smart Crop Prediction System.

1. Soil Sensors (Input Stage):

- The Soil Moisture Meter and Soil Temperature Sensor are connected to the Arduino Nano.
- These sensors collect real-time data about soil moisture and soil temperature.

2. Arduino Processing:

- The Arduino Nano reads the sensor values and displays them on the Arduino IDE Serial Monitor.
- These soil parameters are then sent to the Frontend Interface for further use.

3. Frontend Interface:

- The user interacts with the system through this interface.
- It collects three types of inputs:
 - **From Arduino:** Soil temperature and moisture manually entered the output from the Arduino serial monitor.
 - **From OpenWeatherMap API:** Air temperature, humidity, and average rainfall.
 - **Manual Entry:** Soil nutrients — pH value, Nitrogen, Potassium, and Sodium.

4. Data Flow to Machine Learning Model:

- All the collected input values (sensor data, API data, and manual entries) are sent to the ML Model.
- The model processes these inputs to predict the most suitable crop for the given conditions.

5. Prediction Output:

- The predicted crop is sent back from the ML Model to the Frontend Interface.
- The result is displayed to the user, completing the cycle.

6. EXPERIMENTAL SETUP

The Crop Recommendation System combines both hardware and software components to predict the most suitable crop for cultivation based on soil and environmental data.

6.1 SOFTWARE SETUP:

The backend is developed in Python, with a Machine Learning model trained on the Kaggle Crop Recommendation Dataset. The dataset includes nine key parameters — N, P, K, pH, soil temperature, soil moisture, rainfall, humidity, and ambient temperature. Soil parameters are obtained from sensors, while environmental parameters are fetched dynamically through the OpenWeatherMap API. The web interface, built using Flask, displays the predicted crop to the user.

Link of OpenWeatherMap website: <https://openweathermap.org/>

Multiple algorithms were trained and tested to identify the best-performing model for accurate crop prediction.

Models:

Logistic Regression, Naïve Bayes, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Bagging, AdaBoost, Gradient Boosting, Extra Trees.

Comparison and Justification for Choosing OpenWeatherMap API:

Accurate weather data is essential for effective crop prediction. After comparing major weather APIs like OpenWeatherMap, Weather API, AccuWeather, and Weather Stack, OpenWeatherMap was selected for its strong balance of accuracy, accessibility, and integration ease.

6.2 HARDWARE SETUP:

6.2.1 Equipments used:

6.2.1.1 Microcontroller:

Arduino Nano: The Arduino Nano (Fig 2.) is a compact microcontroller board based on the ATmega328P chip. It offers 14 digital I/O pins, 8 analog inputs, and USB connectivity, making it ideal for small-scale embedded and IoT projects like soil monitoring or crop recommendation systems.

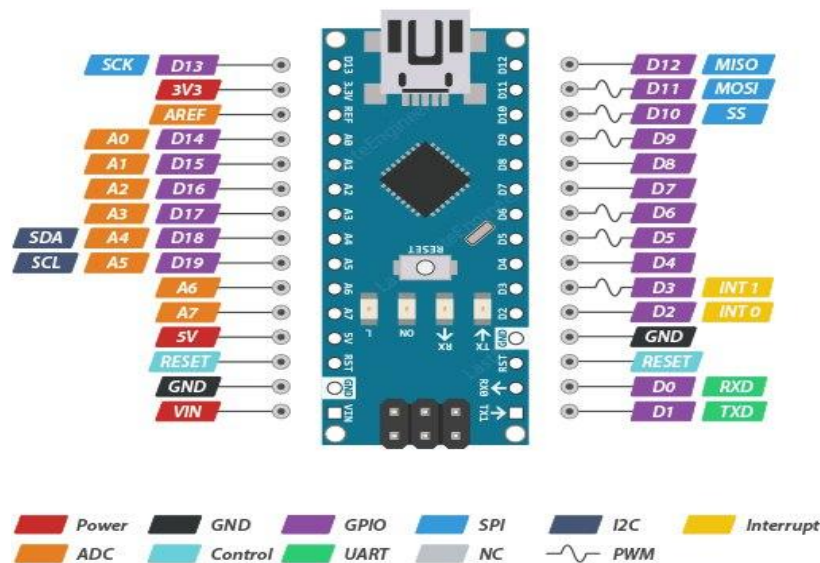


Fig 2. Arduino Nano

6.2.1.2 Sensors:

1. **DS18B20 Temperature Sensor:** The DS18B20 (Fig 3.) is a digital temperature sensor that provides accurate temperature readings from -55°C to $+125^{\circ}\text{C}$. It uses a 1-Wire communication protocol, allowing multiple sensors to connect through a single data pin. The sensor is ideal for soil and environmental temperature monitoring in agricultural applications due to its high precision ($\pm 0.5^{\circ}\text{C}$) and easy digital interface.

Table 2. NPK Sensor Connection to RS485 Module

DS18B20 Pin	Arduino Nano
GND	GND
DATA	D4 (any digital pin)
VCC	5V or 3.3V



Fig 3. DS18B20 Temperature Sensor

2. **Soil Moisture Sensor:** The Soil Moisture Sensor (Fig 4.) measures the water content in the soil to determine how wet or dry it is. It works by detecting changes in electrical resistance or capacitance between its probes. This helps in monitoring irrigation needs and maintaining optimal soil conditions for crop growth.

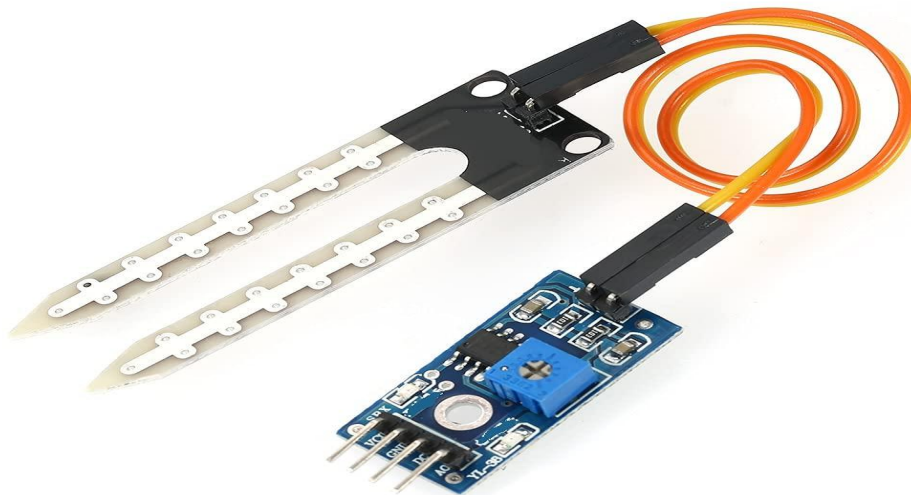


Fig 4. Soil Moisture Sensor

Table 3. Soil Moisture Sensor to Arduino Nano Connection

Soil Moisture Sensor pin	Arduino Nano Pin
GND	GND
VCC	3V3
A0	A0

3. **NPK Sensor:** An NPK sensor (Fig 5.) measures the levels of Nitrogen (N), Phosphorus (P), and Potassium (K) in soil. It helps determine soil fertility and nutrient balance. Farmers use it to apply the right amount of fertilizer for healthier crop growth.



Fig 5. NPK Sensor

Table 4. NPK Sensor Connection to RS485 Module

Wire Colour of NPK Sensor	Connection to RS485 Module
Red	5V
Black	GND
Yellow	RS485-A
Blue	RS485-B

Table 5. RS485 Module to Arduino Nano Connection

RS485 Module Pin	Arduino Nano Pin
RO	D0
DI	D1
RE	D3
DE	D4
VCC	5V
GND	GND

4. **pH Sensor:** A pH sensor (Fig 6.) measures the acidity or alkalinity of a solution by detecting hydrogen ion concentration. It outputs an electrical signal that corresponds to the pH value (0–14 scale). It is commonly used in soil, water quality testing, agriculture, and laboratories.



Fig 6. pH Sensor

Table 6. pH Sensor to Arduino Nano Connection

pH Sensor Pin	Arduino Nano Pin
VCC	5V
GND	GND
PO	A0

6.2.1.3 Other Components:

1. Breadboard
2. 4.7 Kilo Ohm resistor
3. Regulator IC 7805
4. 100 microfarad 25volt capacitor
5. 47 microfarad 25volt capacitor
6. 0.1 microfarad disc capacitor
7. Heat sink (TO220)



Fig 7.1. Breadboard



Fig 7.2. 4.7 Kilo Ohm resistor



Fig 7.3. Regulator IC 7805

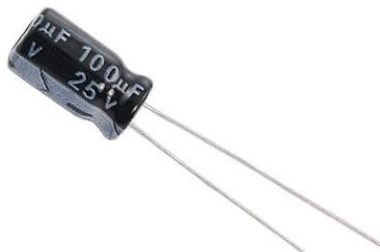


Fig 7.4. 100 microfarad 25volt capacitor

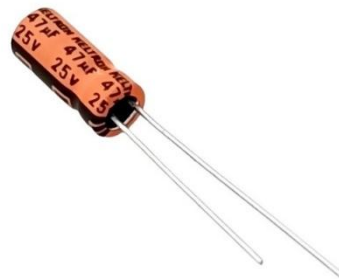


Fig 7.5. 47 microfarad 25volt capacitor



Fig 7.6. Heat Sink



Fig 7.7. 0.1 microfarad disc capacitor

6.2.2 CIRCUIT DIAGRAM:

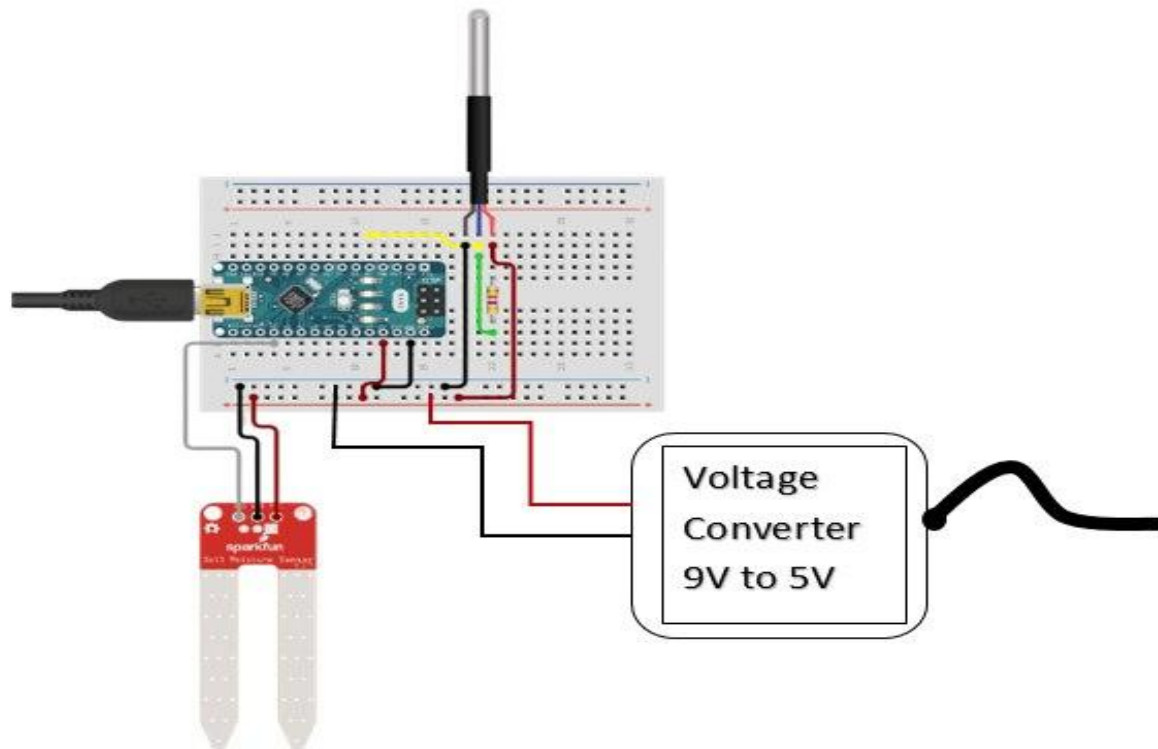


Fig 8. Circuit Diagram

This circuit (Fig 8.) shows a basic IoT-style hardware arrangement where an Arduino board, a soil moisture sensor, and a temperature sensor are connected together on a breadboard and powered through a 9V to 5V voltage converter.

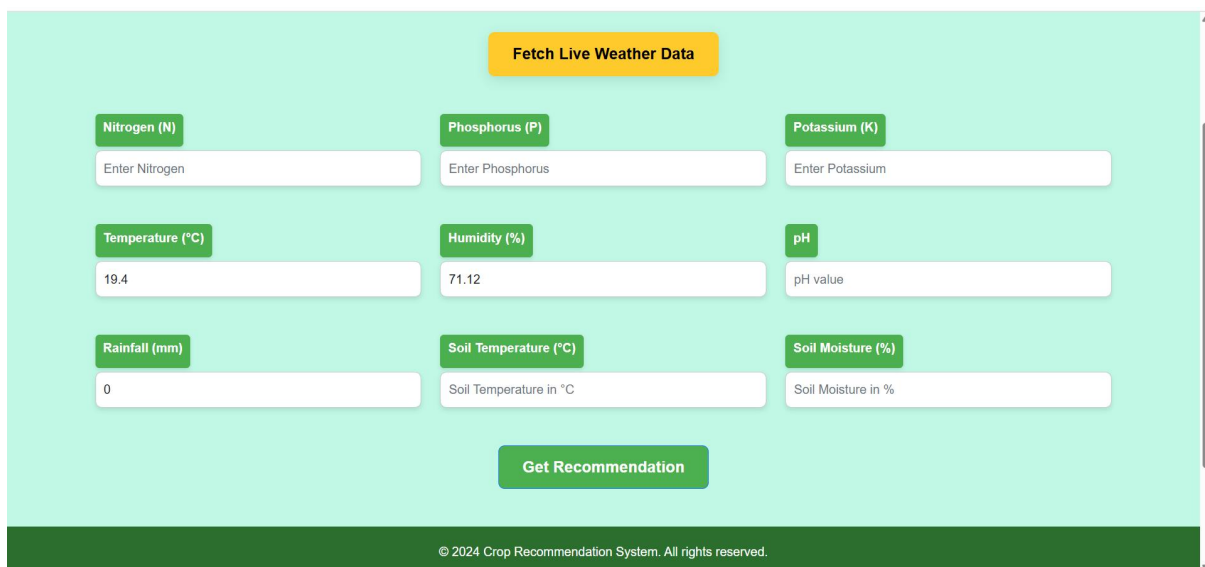
- The Arduino Nano is placed on the breadboard and acts as the main controller.
- A temperature sensor (likely a DS18B20 or similar) is also connected to the Arduino with its power, ground, and data pin.
- The entire setup is powered through a 9V to 5V voltage converter, which takes 9V input (from a battery or adapter) and converts it to a stable 5V required by the Arduino and sensors.
- All components share a common ground, ensuring proper signal reference.
- The Arduino receives sensor values and can process or display them further.
- Other than that we plan on integrating NPK Sensor as well as PH sensor with the microcontroller to achieve complete data about the soil from hardware.

7. RESULTS AND DISCUSSION:

7.1 Software Output:

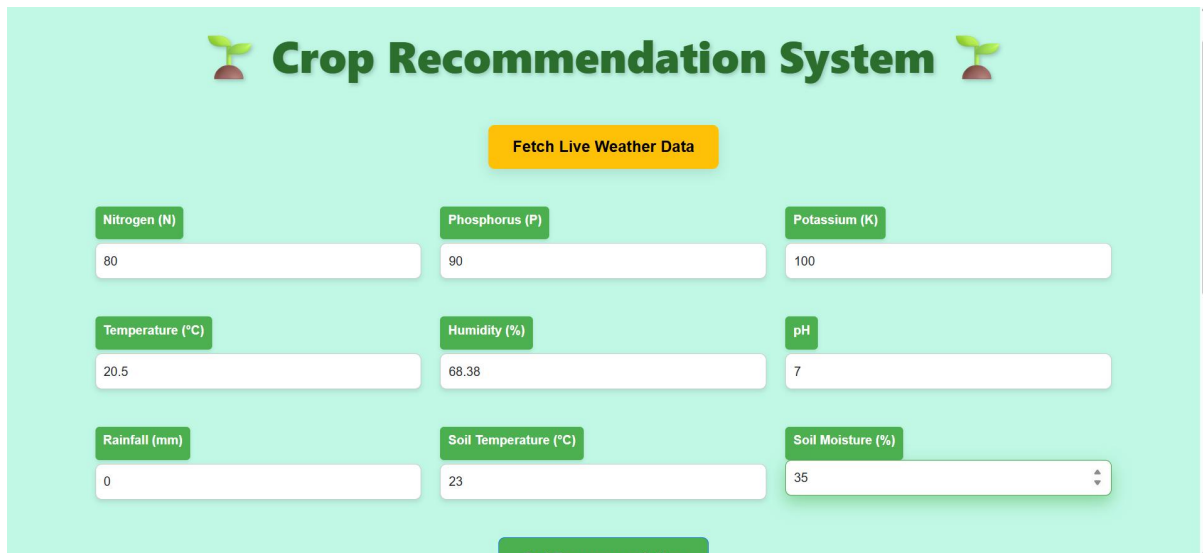
The software module of the Crop Recommendation System successfully processes real-time and user-entered agricultural parameters through the web interface. The interface (Fig 9.1) collects values such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, rainfall, soil temperature, soil moisture, and pH. Once the user clicks on “Get Recommendation”, the backend machine learning model analyzes the input data and generates the most suitable crop for cultivation. It also fetches live weather data and fills the Rainfall, Humidity and Temperature section itself based on the temperature received through the API.

The predicted result is then displayed on the interface along with a visual representation (Fig 9.3) and a short descriptive message, for example: “Banana is the best crop to be cultivated here.” This output confirms the correct functioning of the integrated frontend, backend, and ML model, demonstrating the system’s capability to provide accurate, data-driven crop recommendations based on environmental and soil conditions.



The screenshot displays the input interface of the Crop Recommendation System. At the top, there is a yellow button labeled "Fetch Live Weather Data". Below this, the interface is organized into a grid of input fields for various agricultural parameters. Each field has a green header label and a white input box. The parameters and their current values are: Nitrogen (N) with "Enter Nitrogen", Phosphorus (P) with "Enter Phosphorus", Potassium (K) with "Enter Potassium", Temperature (°C) with "19.4", Humidity (%) with "71.12", pH with "pH value", Rainfall (mm) with "0", Soil Temperature (°C) with "Soil Temperature in °C", and Soil Moisture (%) with "Soil Moisture in %". At the bottom center, there is a green button labeled "Get Recommendation". The footer of the interface contains the text "© 2024 Crop Recommendation System. All rights reserved."

Fig 9.1 Input interface of the Crop Recommendation System displaying fields for entering soil nutrients (N, P, K), temperature, humidity, pH, rainfall, soil temperature, and soil moisture.



Crop Recommendation System

[Fetch Live Weather Data](#)

Nitrogen (N) 80	Phosphorus (P) 90	Potassium (K) 100
Temperature (°C) 20.5	Humidity (%) 68.38	pH 7
Rainfall (mm) 0	Soil Temperature (°C) 23	Soil Moisture (%) 35

[Get Recommendation](#)

Fig 9.2 Updated Crop Recommendation System form with sample values filled in, ready for processing and crop prediction. Live weather data fetched after clicking “Fetch Live Weather Data” button using OpenWeatherMap APIs.

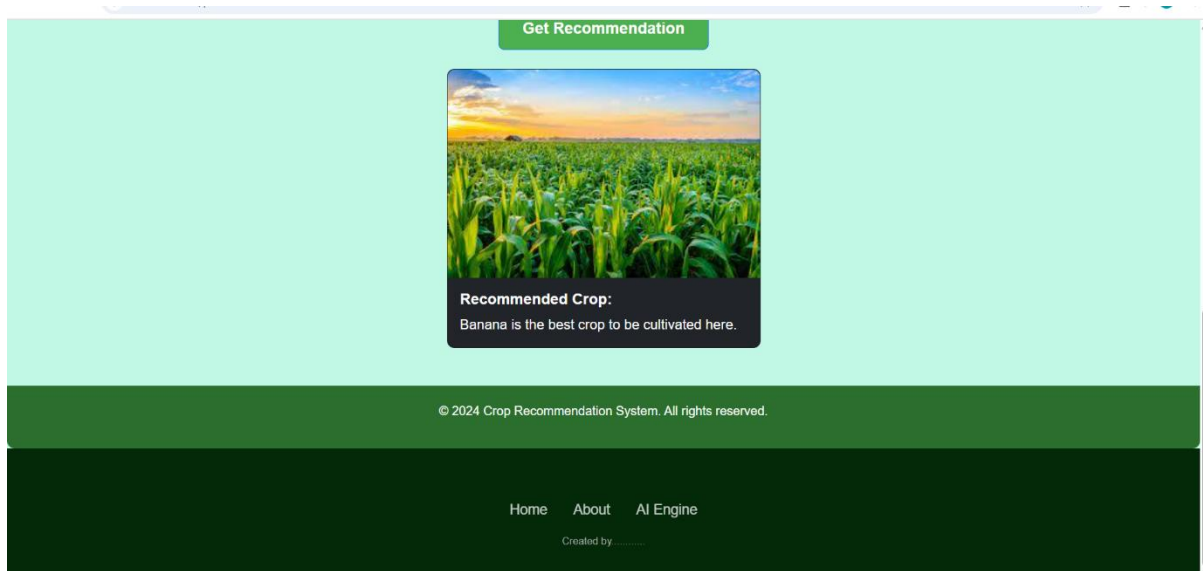


Fig 9.3 Output section of the Crop Recommendation System showing the predicted best crop (Banana) based on the input soil and environmental parameters.

7.2 Hardware Output:

The outputs obtained from the serial monitor (Figs 9.1, 9.2, 9.3), after sensing soil moisture and soil temperature through the connected sensors, will be recorded and manually entered into the website interface. Once these values are submitted, the system will process the input data and display the most suitable or predicted crop based on the sensed soil parameters. This approach ensures accurate prediction and enables users to make informed decisions about crop selection according to real-time environmental and soil conditions.

Output:

```

Soil Moisture: 39%
Temperature: 24.12*C
-----

Soil Moisture: 42%
Temperature: 24.25*C
-----

Soil Moisture: 44%
Temperature: 24.25*C
-----

Soil Moisture: 42%
Temperature: 24.19*C
-----

```

Fig 10.1. Serial
Monitor Output 1

```

Soil Moisture: 58%
Temperature: 24.25*C
-----

Soil Moisture: 58%
Temperature: 24.25*C
-----

Soil Moisture: 61%
Temperature: 24.25*C
-----

Soil Moisture: 52%
Temperature: 24.25*C
-----

Soil Moisture: 13%
Temperature: 24.19*C
-----

```

Fig 10.2. . Serial
Monitor Output 2

```

Soil Moisture: 12%
Temperature: 24.06*C
-----

Soil Moisture: 0%
Temperature: 24.12*C
-----

Soil Moisture: 19%
Temperature: 24.06*C
-----

```

Fig 10.3. . Serial Monitor Output 3

```

Soil Moisture: 21%
Temperature: 24.25*C
-----

```

Fig 10.4. . Serial
Monitor Output 4

8. CONCLUSION

The Smart Crop Prediction System successfully integrates IoT sensors, real-time weather data, and machine learning to provide an efficient and reliable solution for modern agriculture. By collecting soil moisture, soil temperature, environmental factors, and essential nutrient values, the system creates a comprehensive dataset that enables accurate crop recommendations. The seamless flow between hardware, frontend interface, API services, and the ML model ensures that users receive actionable insights instantly.

This project demonstrates how technology can significantly support farmers in making data-driven decisions, improving crop yield, reducing risk, and promoting sustainable farming practices. It highlights the potential of combining IoT and AI to solve real-world agricultural challenges and serves as a strong foundation for future enhancements such as automated irrigation, real-time dashboards, and mobile app integration.

9. ROADMAP OF THE FUTURE WORK

9.1 Proposed Work for the Eighth Semester

In the eighth semester, we plan to organize and integrate all the hardware components in a structured, efficient, and visually presentable manner to ensure smooth functionality and a professional demonstration of the system. Our focus will be on improving both performance and usability through proper hardware alignment and real-time synchronization with the software modules.

As part of hardware enhancement, we will also add N, P, and K nutrient sensors to measure soil Nitrogen, Phosphorus, and Potassium levels directly from the field. This will significantly strengthen the accuracy of the Crop Recommendation System by enabling sensor-based nutrient analysis instead of relying solely on user input.

Furthermore, we intend to integrate a Large Language Model (LLM) into the project, which will serve as an intelligent assistant to help users understand the website's features and functionalities. The LLM will be capable of answering user queries, providing explanations, and guiding them through different sections of the platform. This addition will make the system more interactive, user-centric, and advanced, offering an enhanced experience through natural language communication and AI-driven support.

10. REFERENCES

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